

Shayan Khodabakhsh

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Research Interests

Machine learning for health and behavior sensing: vision-language and temporal models for video and wearable signals, and how measurement errors change downstream feedback decisions. Also EEG-based movement decoding and human-robot shared autonomy.

Education

M.S. in Electrical Engineering | Sep 2025 - expected Fall 2026

University of Rhode Island, Kingston, RI

Thesis in progress: *Multimodal Characterization of Eating Microstructure for Closed-Loop Regulation*. Major advisor: Reza Abiri; co-major advisor: Theodore Walls.

M.S. in Applied Exercise Physiology | Sep 2020 - Oct 2022

Shahid Beheshti University, Tehran, Iran

B.S. in Sports Engineering | Sep 2014 - Sep 2020

Islamic Azad University, Science and Research Branch, Tehran, Iran

Research Experience

Vision-Language Models for Eating Microstructure and Feedback | Jan 2026 - present

Graduate Researcher, University of Rhode Island | Reza Abiri and Theodore Walls | NIH-funded DIBS project

- Developed a video-based bite measurement pipeline using the Qwen2.5-VL visual encoder, LoRA adaptation, and post-encoder temporal attention with past-only, 4-second windows.
- Evaluated 29 participant-held-out laboratory meals with model selection restricted to inner-validation participants; achieved mean bite-event F1 of **0.887**, onset MAE of **382 ms**, and 60-second bite-rate MAE of **0.43 bites/min** (1.63 for a no-video baseline).
- Compared temporal ablations and alternative visual encoders; assessed how missed, additional, and delayed detections affect eating-rate estimates and feedback decisions.
- Evaluated feedback rules through recorded-meal replay and closed-loop simulation. Extending the thesis toward jaw/wrist sensing and multimodal fusion; prospective intervention evaluation remains future work.

Synchronized EEG and Motion Capture for Movement Decoding | Sep 2025 - present

Graduate Researcher, University of Rhode Island | GNN_MOCAP_TNLAB project

- Developed a synchronized acquisition platform integrating **32-channel g.tec EEG**, OptiTrack Motive, and StretchSense glove kinematics using hardware TTL triggers.
- Implemented experimenter and participant interfaces, counterbalanced finger/grasp tasks, impedance monitoring, and trial-level timing and signal-quality checks.
- Built cross-modal alignment and pilot decoding analyses using low-frequency EEG, lagged regression, held-out trials, and artifact/null controls to investigate finger kinematics.

Tripolar Concentric Ring Electrode EEG | Jan 2026 - present

Graduate Researcher, University of Rhode Island | Walter G. Besio

- Led the Python analysis pipeline for a pilot comparison of saline-soaked felt, conductive gel, and paste electrode constructions, using harmonized preprocessing and subject-level statistics.
- Compared alpha signal-to-noise ratio, eyes-open/eyes-closed reactivity, and spectrogram similarity; conducted preprocessing sensitivity analyses and visual-evoked-potential diagnostics.
- First-author methods manuscript in preparation; reproducible analysis package available on GitHub.

Publications and Manuscripts

Peer-Reviewed Journal Articles

1. Chapman, R. M., Taylor, K., Kaczynski, E., **Khodabakhsh, S.**, Richard, S., Hutchinson, J. B., & Marchand, R. C. (2025). Evaluating a commercially available wearable sensor system and its associated calibration procedures for monitoring sagittal knee motion in patients undergoing total knee arthroplasty. *The Knee*, 54, 316-328. doi:10.1016/j.knee.2025.03.001
2. Marchand, R. C., Taylor, K. B., Kaczynski, E. C., Richards, S., Hutchinson, J. B., **Khodabakhsh, S.**, & Chapman, R. M. (2025). Consistency Is Key: A Secondary Analysis of Wearable Motion Sensor Accuracy Measuring Knee Angles Across Activities of Daily Living Before and After Knee Arthroplasty. *Sensors*, 25(13), 3942. doi:10.3390/s25133942

Peer-Reviewed Workshop Paper

1. Arvonen, C., Akinkulore, A., Baik, E., Benavides, E., **Khodabakhsh, S.**, Melanson, K., Walls, T. A., & Thomaz, E. (2026). Demonstration of a Jaw-Mounted Inertial Sensor for Eating Behavior Microstructure Characterization. *Proceedings of the 1st Workshop on Hot Topics in Mobile and Wireless Systems for Health (MobiDx '26)*. doi:10.1145/3842657.3842790

Submitted Manuscripts / Under Review

1. **Khodabakhsh, S.**, Walls, T. A., Arvonen, C. J., Akinkulore, A. J., Thomaz, E., Melanson, K. J., & Abiri, R. (2026). Real-time Vision-Language Models for Behavioral Feedback: A Simulated Closed-Loop Study of Eating Rate. Submitted to *IEEE Journal of Biomedical and Health Informatics* (special issue on AI-based dietary assessment).
2. Farhadi, M. H., Rabiee, A., Ghafoori, S., **Khodabakhsh, S.**, Cetera, A., Fisher, A., Lin, D., & Abiri, R. (2026). End-to-end Optimization of Belief and Policy Learning in Shared Autonomy Paradigms (**BRACE**). *ACM Transactions on Human-Robot Interaction*. Revised manuscript under review.
3. *When to Ask, What to Offer, When to Act: Learning to Assist Users Who Cannot Say What They Want (PRIME)*. Coauthored manuscript submitted to the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027).

Manuscripts in Preparation

1. **Khodabakhsh, S.**, Adeli, B., Norouzi, M., & Besio, W. G. (2026). *A Methods-Harmonized Comparison of Tripolar Concentric Ring Electrode Constructions for Non-Invasive EEG: Saline-Soaked Felt, Conductive Gel, and Paste*.

Additional Research Experience

PRIME: Perception for Human-Robot Shared Autonomy | Jan - May 2026

Graduate Researcher, University of Rhode Island | Reza Abiri

- Fine-tuned YOLO-based perception on task-specific images and integrated object detection with a collaborative robot stack; contributed to end-to-end task evaluation experiments.

Computer Vision for Neuromuscular Assessment | Sep 2025 - present

Graduate Researcher, University of Rhode Island

- Developing markerless hand-pose analysis and vision-based assessment methods for socially assistive robotics and neuromuscular rehabilitation research.

Wearable Sensor Validation | Jun - Sep 2023

Graduate Researcher, University of Rhode Island | R. M. Chapman

- Evaluated IMU-based sagittal knee-angle measurements against optical motion capture and analyzed calibration and sensor-placement effects (two peer-reviewed papers above).

Posters and Conference Abstracts

Research Poster

Khodabakhsh, S., Abiri, R., Akinkurolere, A., Arvonon, C., Melanson, K., Thomaz, E., & Walls, T. (2026). *A Vision Language Coach for Ingestive Behaviors*. ECBE Graduate Student Poster Competition, University of Rhode Island.

Peer-Reviewed Conference Abstracts (No Proceedings)

- **Khodabakhsh, S.**, Marchand, R. C., & Chapman, R. M. (Feb 2024). Nothing but a knee thing: Wearable motion sensors versus optical motion capture measuring knee range of motion before and after TKA. *Orthopaedic Research Society Conference*.
- **Khodabakhsh, S.**, & Chapman, R. M. (Feb 2024). Don't sweat the technique: Accurately measuring knee range of motion using poorly placed wearable sensors. *Orthopaedic Research Society Conference*.

Selected Software Projects

YCB Object Segmentation for Robotic Manipulation | Feb 2026

Fine-tuned YOLOv26n-seg for YCB object segmentation in images and video to support robotic perception. [GitHub](#)

EEG Motor-Intention Classification | Jan 2026

Implemented filter bank common spatial patterns, mutual-information feature selection, and linear discriminant analysis for grasp-type classification. [GitHub](#)

3D Hand Pose Estimation from RGB | Jan 2026

Compared a custom CNN with a pretrained ResNet50 for regression of 21 hand joints using FreiHAND data and cross-validation. [GitHub](#)

Technical Skills

Programming: Python, MATLAB, R.

Machine learning and vision: PyTorch, Keras, scikit-learn, OpenCV; LoRA adaptation, temporal attention, EEG decoding, pose estimation, object detection and segmentation.

Signals, simulation, and acquisition: MNE, PyBullet, g.tec EEG, OptiTrack Motive, Qualysis, StretchSense; EEG/EMG processing and hardware synchronization.

Languages: Persian (native), English (full proficiency).

Honors and Awards

- Dr. Thomas "Doc" Manfredi Student Research Fund, University of Rhode Island (May 2024; Oct 2023).
- Dean's Scholarship, University of Rhode Island (2023).
- Top-five cumulative GPA ranking in master's program, Shahid Beheshti University (2022).

Training

Certificates: Fundamentals of Deep Learning, NVIDIA/URI (Feb 2024); MATLAB Fundamentals, MathWorks/URI (Jan 2023).